

Types of accounting ratio

In theory, it would be possible to calculate literally hundreds of accounting ratios from a single set of financial statements. Some of these ratios would not mean very much but many of them might serve a useful purpose. In practice, however, it is usually sufficient to calculate a fairly small number of key ratios and each of these ratios is described in this chapter. They can be classified into four main groups:

- (a) **Profitability ratios.** The profitability ratios are used to assess whether the business has succeeded in making an acceptable level of profit.
- (b) **Liquidity ratios.** The liquidity ratios are a measure of the ability of the business to pay its debts as they fall due.
- (c) **Efficiency ratios.** The efficiency ratios provide some indication of the extent to which the assets of the business have been efficiently used and managed.
- (d) **Investment ratios.** The investment ratios are mainly of interest to investors or potential investors and may help these users to decide whether or not a business represents a worthwhile investment. However, some of the investment ratios may be of interest to other user groups.

Many of the accounting ratios could be calculated for any kind of business entity (sole trader, partnership or company) but some of them are relevant only to companies. Since ratio analysis is most frequently applied to the financial statements of companies, it is assumed for the remainder of this chapter that the business entity for which ratios are being calculated is a company.

The main ratios in each of the above groups are defined and explained below. Towards the end of the chapter there is a comprehensive numerical example which illustrates the way in which each ratio is calculated.

Profitability ratios

The main profitability ratios[†] are:

- (a) Return on capital employed
- (b) Return on equity
- (c) Gross profit margin
- (d) Net profit margin.

[†] *All of the profitability ratios are illustrated in the comprehensive example (Quandock plc) which is provided later in this chapter.*

Return on capital employed (ROCE)

This important ratio expresses a company's profit as a percentage of the amount of capital invested in the company. However, there are several different ways of calculating ROCE, each using a different definition of the words "profit" and "capital employed". A fairly common definition is as follows:

$$\text{ROCE} = \frac{\text{Profit before long-term interest and tax}}{\text{Share capital and reserves plus non-current liabilities}} \times 100\%$$

This version of ROCE interprets "capital employed" as the *total* amount of money invested in the company in the long term, regardless of whether that money has been supplied by shareholders or lenders. This amount is then compared with the return achieved on that capital. Note that:

- (a) The numerator used in the calculation of ROCE is often referred to as PBIT. If a company's financial statements do not distinguish between short-term and long-term interest, it may be necessary to use the figure for profit before total interest and tax. This will make little difference if the amount of short-term interest is not material.
- (b) Since the denominator used in the calculation of ROCE is equal to the total assets of the company less its current liabilities, this ratio is sometimes referred to as the *return on net assets*.
- (c) Ideally, any ROCE calculation should be based upon the *average* amount of capital employed in the company during the period and this can be approximated by averaging the opening and closing capital figures. However, the calculation of ROCE is often based upon closing capital only, since opening capital may not be known for all of the accounting periods under review. The same point also applies to the calculation of a company's return on equity (see below).
- (d) If a company's capital employed is understated in the financial statements, possibly because non-current assets such as property are shown at cost rather than current value, the ROCE figure will be misleadingly high. This point highlights the fact that accounting ratios are affected by accounting policies (see later in this chapter).

As stated above, ROCE can be defined as PBIT divided by net assets (where net assets are defined for this purpose as total assets less current liabilities). This view of ROCE makes it clear that the ROCE ratio can actually be broken down into two subsidiary components, as indicated by the following formula:

$$\text{ROCE} = \frac{\text{PBIT}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Net assets}}$$

The first of these components is a profitability ratio known as the "profit margin" and the second component is an efficiency ratio known as "asset turnover". Both of these ratios are explained in more detail later in this chapter. Breaking down ROCE into its component parts may help to explain why a company's ROCE has changed from one period to another or why one company's ROCE is different from another's.

EXAMPLE 1

The following information has been extracted from the financial statements of two companies for the year to 31 December 2019. The companies are of similar size and operate in the same sector of industry.

	Company A	Company B
	£000	£000
Sales	14,400	13,800
Profit before long-term interest and tax	720	690
Share capital and reserves	3,100	2,475
Non-current liabilities	500	400

Calculate ROCE for each company and analyse each company's ROCE figure into its component parts.

Solution

	<i>ROCE</i>	=	<i>Profit margin</i>	×	<i>Asset turnover</i>
Company A	$\frac{720}{3,600}$	=	$\frac{720}{14,400}$	×	$\frac{14,400}{3,600}$
	20%	=	5%	×	4.0
Company B	$\frac{690}{2,875}$	=	$\frac{690}{13,800}$	×	$\frac{13,800}{2,875}$
	24%	=	5%	×	4.8

These calculations show that Company B has the same profit margin as Company A but has a better ROCE because of its better asset turnover.

Return on equity (ROE)

This profitability ratio concentrates on the company's ordinary shareholders and compares their capital with the amount of profit which has been earned on their behalf. Return on equity is usually calculated as follows:

$$\text{ROE} = \frac{\text{Profit after interest, tax and preference dividend}}{\text{Ordinary share capital and reserves}} \times 100\%$$

This ratio may also be classified as an investment ratio, since it can be used by investors to assess the desirability of a company's ordinary shares. Investment ratios are discussed in more detail later in this chapter.

As well as comparing a company's ROCE and ROE for an accounting period with the returns achieved in previous periods and by other companies, it may also be useful to compare these ratios with current market rates of interest. For instance, if a company's ROCE is lower than current borrowing rates, this might indicate that the company would

not benefit from making further borrowings. Similarly, if a company's ROE is higher than current savings rates, this might indicate that the company's ordinary shares are a good investment. However, such comparisons should be viewed with caution. Returns in future accounting periods may be very different from those in past periods and factors such as the degree of risk associated with various investments should also be taken into account.

Gross profit margin

The gross profit margin expresses a company's gross profit as a percentage of its sales revenue and is sometimes referred to as the *gross profit percentage*. Gross profit margin is calculated as follows:

$$\text{Gross profit margin} = \frac{\text{Gross profit}}{\text{Sales}} \times 100\%$$

It would be very misleading to compare the gross profit margin of two totally dissimilar businesses (e.g. a jeweller and a supermarket) but a comparison between two companies in the same line of trade should be meaningful and might throw some light on the pricing policies adopted by each of the companies concerned.

Net profit margin

The net profit margin (also referred to as simply the *profit margin*) expresses a company's profit as a percentage of its sales revenue and is calculated as follows:

$$\text{Net profit margin} = \frac{\text{Profit}}{\text{Sales}} \times 100\%$$

However, there is no consensus as to which profit figure should be used in the calculation of this ratio. Possibilities include PBIT, profit before tax and profit after tax. Therefore it is very important to ascertain the way in which the ratio has been calculated before basing important decisions upon it.

A year-on-year comparison of a company's gross profit margin and net profit margin may reveal the extent to which operating expenses are under control. For instance, if net profit margin has increased whilst gross profit margin has remained static, this might indicate that the company has succeeded in trimming its operating expenses. However, the validity of this conclusion will depend upon the way in which the net profit margin has been calculated and the extent to which the company's profit has been affected by items such as "other income" (see Chapter 3).